THANDEKA NGETU

EXERCISE 2: SQL AGGREGATE & OPERATORS

Q1. SELECCT DISTINCT department
FROM students;

department
IT
HR
Finance

Q2. Get the average age of students per department.

SELECT department,
AVG (age) AS avg_age
FROM students
GROUP BY departments;

department	avg_age
IT	20,5
HR	22
Finance	23

03 Show departments with more than 1 student

```
SELECT department,
COUNT (student_id) AS student_count
FROM students
GROUP BY department
HAVING COUNT (student_id) > 1;
```

department	student_count
IT	2
HR	2

04. Get all students whose age between 21 and 23

```
SELECT (*)
FROM students
WHERE age BETWEEN 21 AND 23;
```

student_id	name	age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

05. List all students in the IT or HR department who are older than 21

```
SELECT (*)
FROM students
WHERE department IN ('IT', 'HR')
AND age > 21;
```

student_id	name	age	department
2	Bob	22	HR
5	Eve	22	HR

06. Show total credits per department, only for departments with more than 5 total credits.

```
SELECT department,
       SUM (credits) AS total_credits
FROM courses
GROUP BY department
HAVING SUM (credits) > 5;
```

department	total_credits
IT	11

07. List all courses that do not have 4 credits

```
SELECT (*)
FROM courses
WHERE credit <> 4;
```

course_id	course_name	department	credits
101	SQL Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

08. Show the top 3 courses by credits in descending order.

```
SELECT course_id,
       course_name,
       credits
FROM courses
ORDER BY credits DESC
LIMIT 3;
```

course_id	course_name	credits
101	SQL Basic	3
102	Python	4
103	Data Science	4

09. Get the MAX, MIN & AVG grade across all enrollments

```
SELECT MAX(grade) AS max_gradede,
       MIN(grade) AS min_grade,
       AVG(grade) AS avg_grade
FROM enrollments;
```

max_grade	min_grade	avg_grade
90	78	84,6

10. Count how many enrollments exist per course

```
SELECT course_id,
       COUNT(enrollment_id) AS enrollment_count
FROM enrollments
GROUP BY course_id;
```

course_id	enrollment_count
101	1
102	1
103	1
104	1
105	1

11. Find total salary and total bonus per department

```
SELECT department,
       SUM(salary) AS total_salary,
       SUM(bonus) AS total_bonus
FROM salaries
GROUP BY department;
```

department	total_salary	total_bonus
IT	122 000	10 500
HR	109 000	7 500
Finance	70 000	6 000

12. Show departments where average salary is above 55 000.

```
SELECT department,
       AVG(salary) AS avg_salary
FROM salaries
GROUP BY department
HAVING AVG (salary) > 55000;
```

department	avg_salary
IT	61 000
Finance	70 000

13. List employe
60 000.

SELECT

FROM
WHE

emplo

13. List employees whose salary plus bonus is greater than 60 000.

```
SELECT employee_id,
       name,
       salary,
       bonus
       salary + bonus  AS total_compensation
FROM salaries
WHERE salary + bonus > 60000;
```

employee_id	name	salary	bonus	total_compensation
1	TOM	60 000	5 000	65 000
3	Spike	70 000	6 000	76 000
4	Tyke	62 000	5500	67 500

14. Show total & average budget per department. Only include departments with average budget above 70 000

```
SELECT department
       SUM( budget) AS total_budget,
       AVG (budget) AS 'avg_budget
FROM projects
GROUP BY department
HAVING AVG (budget) > 70000;
```

department	total_budget	avg_budget
IT	270 000	135 000
Finance	80 000	80 000

15. List all projects with budgets between 50 000 & 120 000 excluding the marketing department.

```
SELECT project_id,
       project_name,
       department,
       budgect
FROM projects
WHERE budget BETWEEN 50000 AND 120000
AND department <> 'Marketing';
```

project_id	projet_name	department	budget
1	AI App	IT	120 000
2	Payroll System	Finance	80 000
4	Website	Marketing	60 000
5	HR Portal	HR	50 000